

INCLINED PLANE: THE MAGIC SLIDE



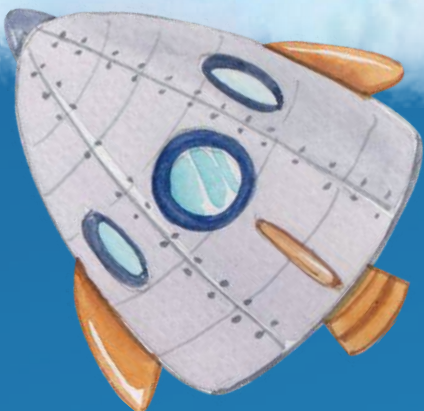


THE PLAYGROUND SLIDE HOOK

Why can you slide down smoothly
but not climb the wall?

Inclined Plane = slanting helper 🎉





WHAT IS AN INCLINED PLANE?

A ramp is a flat slanting surface
that helps push heavy things.
See how the plank makes moving
easier!



STEEP VS. GENTLE SLOPES

Steep slope: hard but short. Gentle slope: easy but long.



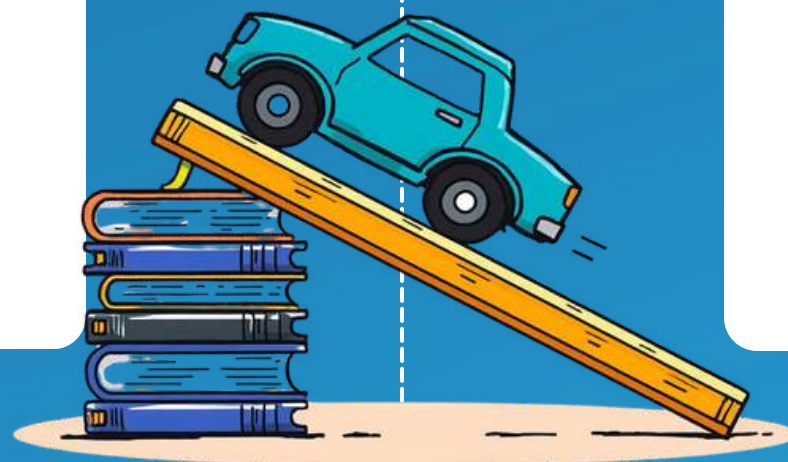
● Steep Slope

- ⌘ Lots of effort needed
- ⌘ Short distance to climb
- ⌘ Like a steep hill road in Shimla
- ⌘ Hard on your legs!



● Gentle Slope

- ⌘ Less effort needed
- ⌘ Longer path to walk
- ⌘ Like a winding temple path
- ⌘ Easy and smooth!



Hospital Ramp

Hospitals have smooth ramps near the entrance so stretchers, wheelchairs, and elders can move in and out safely without climbing stairs.

Home Ramp

Many homes and apartments have ramps at the gate so bicycles, trolleys, and grandparents can enter without stepping over high edges.

SPOTTING THE
SLANT IN REAL LIFE!





Metro stations have smooth ramps for luggage trolleys and wheelchairs. Everyone can move easily from platform to exit!



Big railway stations like Howrah and New Delhi have gentle ramps so passengers with heavy bags can walk up without stairs.



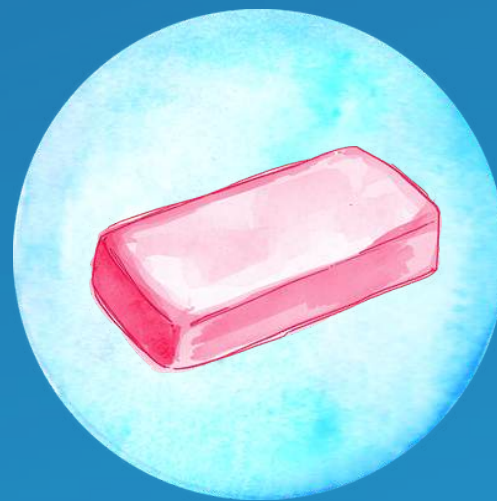
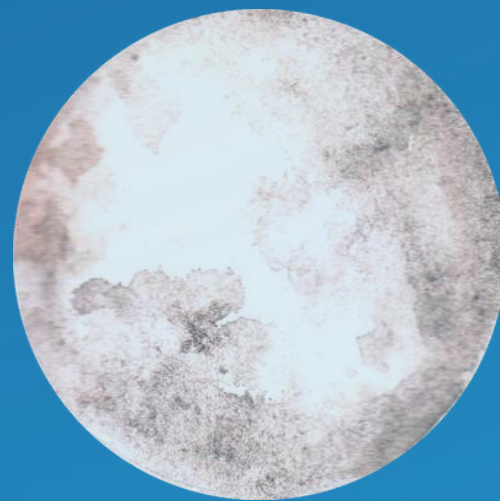
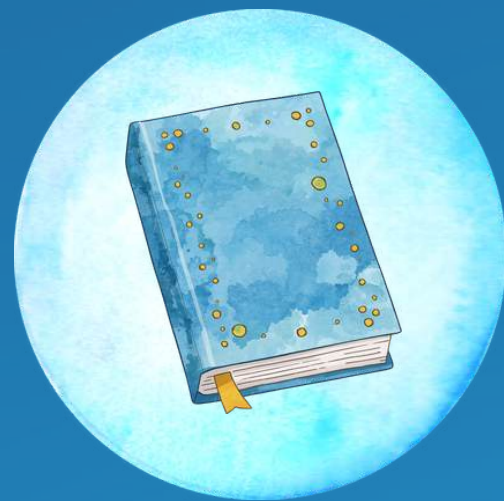
Hospital entrances have wide ramps for stretchers and elders. The gentle slope makes it safe and easy for everyone.

MEET THE RAMP HELPERS!



HANDS-ON MICRO LAB: BUILD YOUR RAMP!

Build a ramp using your textbook and eraser! Stack one book, lean the scale to form a ramp, and get ready to explore the inclined plane!



HANDS-ON LAB: LET'S EXPERIMENT!

Step 1

Put the eraser on the ramp. Does it slide down? Watch what happens — friction and gravity are playing tug of war!



Step 2

Now put the toy car on the ramp. Watch it roll down fast! The wheels beat friction — zoom!

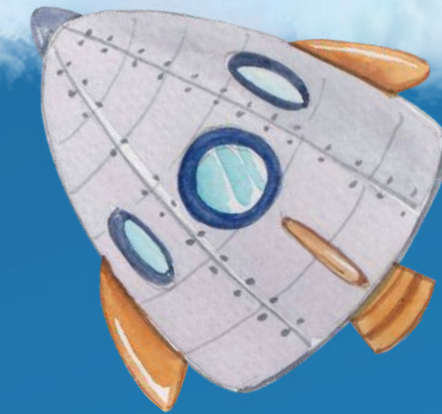




STEP 3: MAKE IT STEEPER!

Add a second book under the ramp.
What happens now?
The toy car rolls down even faster!
Steeper slope = more speed!





STORY TIME: THE SMART ELEPHANT GAJRAJ

Gajraj the elephant rolls heavy logs up a sloping mound to save energy. Instead of lifting, he uses the ramp!

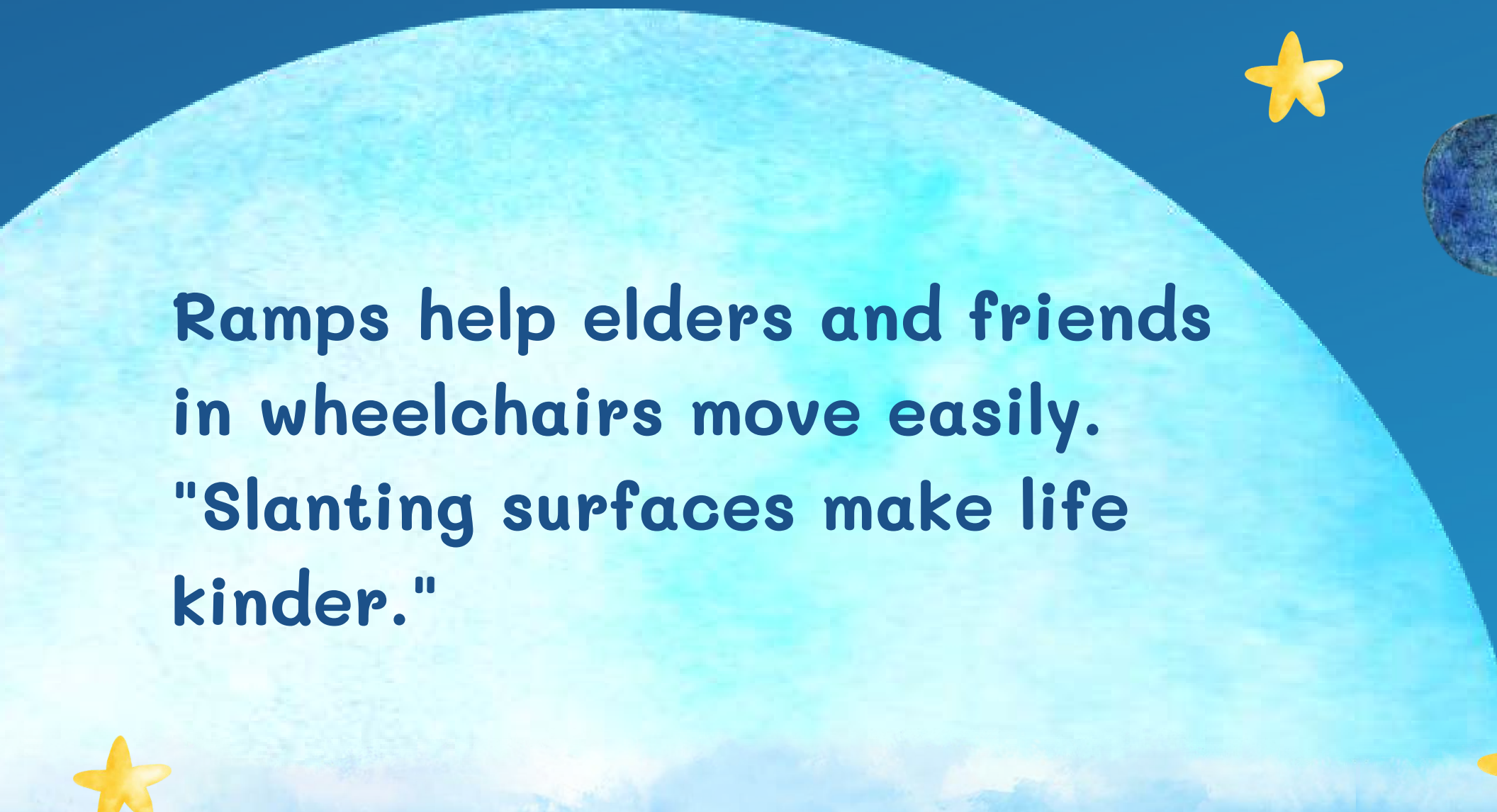
"Work smarter, not harder!"

The inclined plane is Gajraj's secret helper.





WHY RAMPS MATTER FOR EVERYONE



Ramps help elders and friends
in wheelchairs move easily.
"Slanting surfaces make life
kinder."



RECAP: THE MAGIC OF INCLINED PLANES

Ramp = Slanting Helper

A ramp is a flat, slanting surface that makes it easier to move heavy things up or down. It is one of the simplest machines!

Gentle Slope is Easier

A gentle slope needs less effort to push things up, but the path is longer. A steep slope is shorter but much harder!

Ramps Help Everyone Move Safely

Ramps help elders, wheelchair users, and even luggage trolleys move safely. They make life easier and kinder for everyone!



**GREAT JOB TODAY!
SEE YOU NEXT TIME!**

 **Weekly Mission: Find a ramp near your home or school.
Draw it or photograph it and share with your class!**

